

# Analyzing the Effectiveness of Adjuvant Therapy for Colon Cancer using R Language

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**Abstract:** Colorectal cancer claimed 774,000 lives in the year 2015 according to World Health Organization. The treatment after surgery with or without chemotherapy may offer relief from the disease for a few patients. A well-known dataset was taken as input to an experimental model in this paper. Using R language for implementing the model, the results are generated and some of them are rendered in a graphical manner. The analysis of results and the conclusions are presented in such a way that they might offer useful insight into understanding which risk factors should be closely monitored in people who are at risk for colon cancer.

**Keywords:** colon, cancer, R, regression, statistics.

## I. INTRODUCTION

A cancer is an abnormal growth of cells that could cause potentially fatal health issues. Colon shown in Figure 1 is the last part of the digestive system [1]. Colon cancer is categorized under colorectal cancer. According to the World Health Organization's report [2], colorectal cancer is the third-most common form of cancer. It accounts for ten percent of all cancer cases diagnosed world-wide.

Adjuvant therapy means the therapy that is administered after surgery to suppress secondary tumor formation.

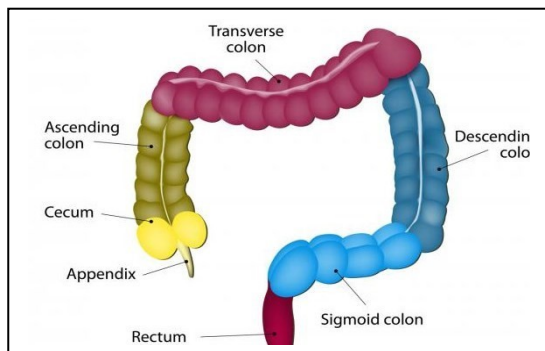


Figure 1. Anatomy of the Large Intestine

An analysis of the colon cancer data available from the patients admitted in hospitals for chemotherapy treatment would prove invaluable in handling similar cases in the future.

Section II of this paper provides an overview of the dataset. Section III presents a brief overview of R programming language and its features. Section IV proposes an experimental model for analyzing the dataset. Section V explains the results in order to facilitate an understanding of which risk factors are dominant in colon cancer tumor

formation and death rate.

## II. DATASET DESCRIPTION

The dataset used in this paper is obtained from [3]. It was described in [4] and [5]. It pertains to a successful trial of adjuvant chemotherapy treatment on patients suffering from stage B and stage C of colon cancer [6]. The study was conducted at the North Central Cancer Treatment Group and the Mayo Clinic. They were randomly assigned to three groups:

- Group 0: Under observation with no further therapy.
- Group 1: Treatment with Levamisole drug.
- Group 2: Treatment with the combination of Levamisole and Fluorouracil drugs.

The description of each field in the dataset is as follows:

1. **id:** Patient ID.
2. **study:** Study ID is set to 1 for all patients.
3. **rx:** Type of treatment.
  - Obs = Observation [ group 0 ]
  - Lev = Levamisole [ group 1 ]
  - Lev+5-FU = Levamisole+5-FU [ group 2 ]
- When Levamisole is used in combination with another drug named Fluorouracil, the combination is referred to as "Levamisole+5-FU" or "Lev+5-FU".
4. **sex:** Gender. 1 for male patients. 0 for female patients.
5. **age:** Age of the patient in years.
6. **obstruct:** If the colon is obstructed by the tumor, this field is set to 1. Otherwise, it is set to 0.
7. **perfor:** If the colon is perforated due to the cancer, this field is set to 1. Otherwise, it is set to 0.
8. **adhere:** This field is set to 1 if the cancer has spread to other organs near the colon.
9. **nodes:** Number of lymph nodes in the colorectal region with detectable cancer. This field is related to the node4 field.
10. **time:** Days until event or censoring. Censoring means removing the patient from the study sample.
11. **status:** Censoring status. This field is set to 1 if the patient is removed from sample.
12. **differ:** Differentiation of tumor. 1=well, 2=moderate, 3=poor.
13. **extent:** Extent of local spread of the cancer. 1=submucosa, 2=muscle, 3=serosa, 4=contiguous structures.
14. **surg:** Time from surgery to registration. 0=short, 1=long.
15. **node4:** This field is set to 1 if the cancer had spread to more than four lymph nodes. Otherwise, it is set to 0.
16. **etype:** Event type. This field is set to 1 if the patient suffered recurrence of cancer and set to 2 if the patient died.

The patient characteristics are summarized in Table 1.

Table 1. Patient Characteristics

| Patient Characteristics                         |          |
|-------------------------------------------------|----------|
| Male                                            | 484      |
| Female                                          | 445      |
| Age: Median                                     | 61       |
| Age: Range                                      | 18 to 85 |
| Patients who received no treatment              | 315      |
| Patients given Levamisole drug                  | 310      |
| Patients given Levamisole+5-FU drug combination | 304      |

### III. R LANGUAGE

R is an open-source object-oriented programming language that is supported by the R Foundation for Statistical Computing. It could be used for reading data, manipulating data, analyzing data and depicting data in the form of graphs, charts or plots. The R language is being widely used among statisticians and data miners.

R includes many features that are part of the S language which is commercially available from Bell Laboratories, USA. R's salient features are as follows [7]:

- Support for various types of homogeneous and heterogeneous data structures.
- S3 and S4 classes, which are modeled after the classes available in S language.
- Graphical rendering capability.
- Many packages that are developed and made available for free by the active R developers' community.
- Saving of commands and data between sessions.

### IV. EXPERIMENTAL MODEL

The analysis of data was done using R programming language version 3.3.2 and RStudio integrated development environment (IDE) version 1.1.383 [8].

#### A. Reading the dataset

One of the data structures provided by R language is a data frame. It is a two-dimensional structure that can store heterogeneous data. In order to read the dataset available on a public website, the following command was used in R:

```
colcan <-
read.csv("https://raw.githubusercontent.com/vincentarelbundock/
Rdatasets/master/csv/survival/colon.csv",
header=T)
```

In the above command, *colcan* is a data frame that

consists of all the data read from the website. Header information from the dataset is retained in the data frame. The first few rows of this data frame are displayed in Table 2. The remaining data is not shown here due to space constraints.

Table 2. First Few Rows from the Dataset: Columns 1 to 9

| id | study | rx      | sex | age | obstruc | perfor | adhere | nodes |
|----|-------|---------|-----|-----|---------|--------|--------|-------|
| 1  | 1     | Lev+5FU | 1   | 43  | 0       | 0      | 0      | 5     |
| 1  | 1     | Lev+5FU | 1   | 43  | 0       | 0      | 0      | 5     |
| 2  | 1     | Lev+5FU | 1   | 63  | 0       | 0      | 0      | 1     |
| 2  | 1     | Lev+5FU | 1   | 63  | 0       | 0      | 0      | 1     |
| 3  | 1     | Obs     | 0   | 71  | 0       | 0      | 1      | 7     |
| 3  | 1     | Obs     | 0   | 71  | 0       | 0      | 1      | 7     |
| 4  | 1     | Lev+5FU | 0   | 66  | 1       | 0      | 0      | 6     |
| 4  | 1     | Lev+5FU | 0   | 66  | 1       | 0      | 0      | 6     |

#### B. Cleaning the dataset

In the dataset, the field named *rx* is not a numerical value. It would not be possible to use it for quantitative analysis. Therefore, the field value was changed as follows:

- 0 = Observation
- 1 = Treatment with Levamisole drug
- 2 = Treatment with Levamisole+5-FU drug

#### C. Simple linear regression

*lm()* is a function available in R language for performing simple linear regression on the data available in a data frame. Its syntax is *lm ( dependent variable ~ independent variables, data = data frame name )*. It means that dependent variable is being modeled as a combination of independent variables from the given data frame.

#### D. Multiple linear regression

Predicting a quantitative response *Y* based on the basis of multiple predictor variables  $X_0, X_1, X_2, \dots, X_n$  is known as multiple linear regression [9]. The multiple linear regression is often represented in the form of the equation 1.

$$Y \approx \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n + \varepsilon \quad (1)$$

There are several aspects of the above equation that merit further explanation.

1. *Y* is approximately modeled as a combination of multiple *X* predictor variables.
2. *Y* is being regressed on multiple *X* predictor variables.

3. Coefficients  $\beta_1$  to  $\beta_n$  are the regression coefficients. They are generally unknown and they need to be estimated in such a way that the relationship between X variables and the dependent Y variables can be explained satisfactorily.
4.  $\beta_0$  is the intercept term, i.e., the expected value of Y when X values are equal to 0.
5.  $\beta_1$  is interpreted as the average effect on Y when there is one unit increase in X, holding all other predictor variables as fixed.
6. The error term  $\varepsilon$  represents the value that cannot be explained by the relationship between X variables and Y. It is typically independent of the X variables.

### E. Dataset analysis of survival records

#### Survival length in different groups

Initially, an effort was made to understand if the recurrence-free length of patients depended on the type of treatment received by them. All the records related to patient survival were pulled into a separate data frame known as *colsurvive*:

```
colsurvive <- colcan[ colcan$etype == 1, ]
```

The recurrence-free intervals of group 0, group 1 and group 2 patients are generated using the *hist()* function in R is shown below.

```
group0surv <- colsurvive[ colsurvive$rx == 0, ]
hist( round( group0surv$time/365 ), col="green",
      xlab="Years of survival", ylab="Number of Patients",
      main="Recurrence-free Interval in Group 0" )
```

```
group1surv <- colsurvive[ colsurvive$rx == 1, ]
hist( round( group1surv$time/365 ), col="grey",
      xlab="Years of survival", ylab="Number of Patients",
      main="Recurrence-free Interval in Group 1" )
```

```
group2surv <- colsurvive[ colsurvive$rx == 2, ]
hist( round( group2surv$time/365 ), col="blue",
      xlab="Years of survival", ylab="Number of Patients",
      main="Recurrence-free Interval in Group 2" )
```

Figure 2 shows that the group 2 has the most number of patients in the recurrence-free survival period of 4 to 8 years. The median of recurrence-free survival interval in group 0, group 1 and group 2 is approximately 2.8 years, 2.81 years and 5.56 years respectively. So, the “Lev+5-FU” combination does seem to have a noticeable effect on increasing the number of years for which colon cancer patients could live without any recurrence of the disease.

#### Type of treatment, gender and disease spread to lymph nodes

A multiple linear regression was done to understand whether the number of days of survival depended on the type of treatment, gender and/or the spread of the disease to more than four lymph nodes. The *lm()* function in R language performs linear

regression. The R code used is shown below.

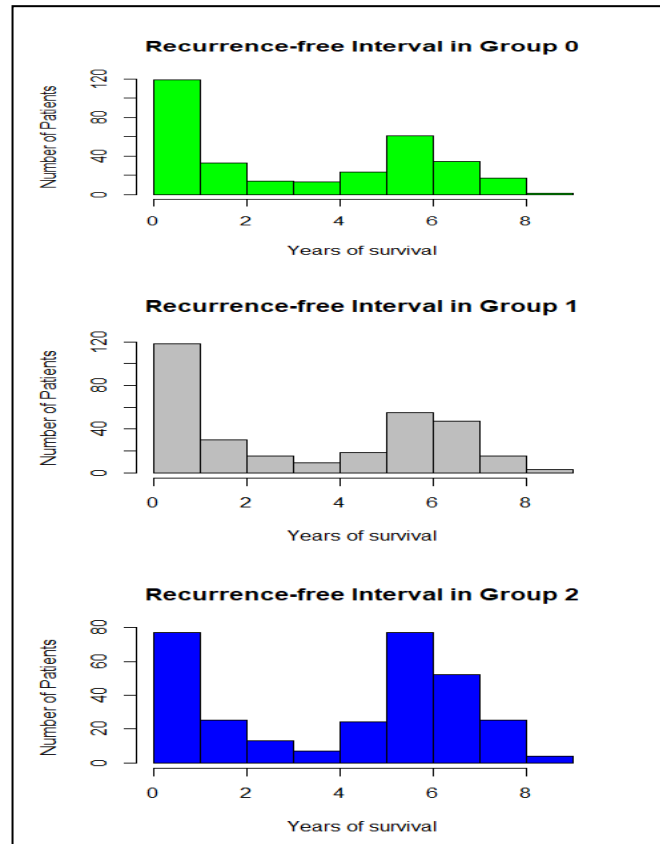


Figure 2. Recurrence-free survival in three groups of patients

```
lm2 <- lm( colsurvive$time ~ colsurvive$rx +
           colsurvive$sex + colsurvive$node4 )
```

The output of the above function call is stored in a frame known as *lm2*. *summary()* function in R language provides the details about linear regression as shown in Figure 3.

In this output, the coefficients section is of primary interest in interpreting the results. A p-value of less than 0.05 is considered significant in statistical analysis.

The type of treatment has substantial impact on the survival length of a patient who was admitted for treatment after surgery. The patients in whom cancer spread to more than four lymph nodes is highly significant and has a direct bearing upon the survival length of those patients.

The intercept is also significant, indicating that the factors other than the three chosen for this multiple linear regression play an important role as well. However, the gender of patients appears to have no impact on the survival length of the patients.

```
summary( lm2 )

Call:
lm(formula = colsurvive$time ~ colsurvive$rx +
    colsurvive$sex + colsurvive$node4)

Residuals:
    Min       1Q   Median       3Q      Max
-1716.71  -851.78   94.29   775.22  2064.29

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)    1403.33     63.40  22.136 < 2e-
16 ***
colsurvive$rx     166.46     38.17   4.362 1.44e-
05 ***
colsurvive$sex     25.45     62.41   0.408  0.683
colsurvive$node4  -640.99     69.78  -9.186 < 2e-
16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.'
0.1 ' ' 1
```

Figure 3. Multiple regression - I

### Extent of disease spread

The extent of disease spread is indicated in terms of whether the cancer had spread to membranes, tissues and muscles surrounding the colorectal area. A simple linear regression was performed by modeling the survival length of a patient on the extent of disease spread to areas surrounding the colon. Note that this is different from the spread of the disease to the lymph nodes discussed earlier.

```
lm3 <- lm( coldeath$time ~ coldeath$extent )
```

The output of the above function call is stored in a frame known as *lm3*. `summary()` function in R language provides the details about linear regression as shown in Figure 4.

From the below output of simple linear regression, it is evident that the extent of the disease spread to areas surrounding the colon is highly significant because p-value is less than 0.05. The intercept is also significant, suggesting that the survival length is determined by other factors too.

### Age of the patient, perforation and obstruction of colon

A multiple linear regression was performed by taking the age of the patient, perforation and obstruction of the colon. The output is shown in Figure 5.

```
lm3 <- lm ( colsurvive$time ~ colsurvive$obstruct +
    colsurvive$perfor + colsurvive$age )
```

```
summary(lm3)

Call:
lm(formula = coldeath$time ~ coldeath$extent)

Residuals:
    Min       1Q   Median       3Q      Max
-1716.1  -827.4   236.9   687.6  1695.6

Coefficients:
            Estimate Std. Error t value Pr
(>|t|)
(Intercept)    2604.61     168.96  15.42 <
2e-16 ***
coldeath$extent -323.75     57.71  -5.61 2.
67e-08 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.
05 '.' 0.1 ' ' 1
```

Figure 4. Simple linear regression

From the output in Figure 5, it can be concluded that age and perforation of the colon do not determine the survival of the patient under observation or undergoing adjuvant chemotherapy. The reason for the intercept having a high value is that the survival could be explained by factors other than age and perforation of the colon.

```
summary( lms3 )

Call:
lm(formula = colsurvive$time ~ colsurvive$obstruct
    + colsurvive$perfor + colsurvive$age)

Residuals:
    Min       1Q   Median       3Q      Max
-1439.61 -1006.68   98.76   881.16  1900.04

Coefficients:
            Estimate Std. Error t value Pr(>
|t|)
(Intercept)    1348.444     170.138   7.926 6.51
e-15 ***
colsurvive$obstruct -167.233     83.404  -2.005  0
.0452 *
colsurvive$perfor  -245.506     195.309  -1.257  0.
2091
colsurvive$age      1.610       2.753   0.585  0.
5587
```

Figure 5. Multiple regression - II

## Hypothesis Testing

Hypothesis testing is a useful statistical technique to ascertain if there is any meaningful correlation between two variables. In this technique, the null hypothesis describes the case where there is no correlation between two variables. The alternative hypothesis describes the case where there is correlation between two variables.

If the p-value turns out to be less than 0.05, then the null hypothesis is rejected. Otherwise, the null hypothesis is accepted. This is the usual way of eliminating the disconfirmed hypothesis.

## Analysis of death records

To understand the impact of adherence to nearby organs on the survival length, a correlation test was performed using Pearson's product-moment correlation as shown in Figure 6. In this test, the null hypothesis is that there is no correlation between adherence to nearby organs and the survival length in days. Since the p-value is less than 0.05, the null hypothesis is rejected. So, the adherence of colon cancer to nearby organs has an impact on the number of days of survival.

```
> cor.test( coldeath$time, coldeath$adhere )

Pearson's product-moment correlation

data:  coldeath$time and coldeath$adhere
t = -2.1968, df = 927, p-value = 0.02828
alternative hypothesis: true correlation is no
t equal to 0
95 percent confidence interval:
 -0.135657805 -0.007682312
sample estimates:
      cor
-0.07196625
```

Figure 6. Correlation test

The correlation is pictorially represented in Figure 7.

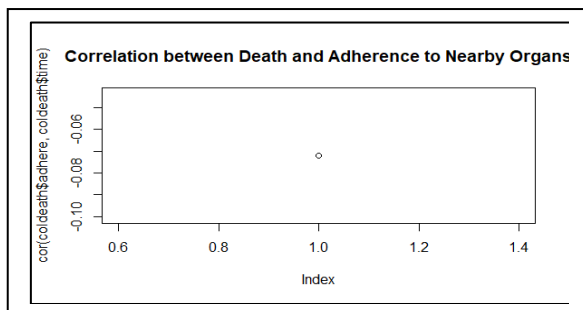


Figure 7. Correlation between adherence and survival length in days

## V. CONCLUSION

In this paper, a dataset pertaining to colorectal cancer patients was analyzed using R programming language. The following conclusions are made based on the analysis of a single trial and also presented in Table 3.

Table 3. Conclusions in tabular manner

| S. No. | Dataset Field | p-value  | Remarks                       |
|--------|---------------|----------|-------------------------------|
| 1.     | age           | 0.5587   | Not a significant risk factor |
| 2.     | perfor        | 0.2091   | Not a significant risk factor |
| 3.     | extent        | 2.67e-08 | Significant risk factor       |
| 4.     | rx            | 1.44e-05 | Significant risk factor       |
| 5.     | sex           | 0.683    | Not a significant risk factor |
| 6.     | node4         | 2e-16    | Significant risk factor       |
| 7.     | obstruct      | 0.0452   | Not a significant risk factor |

1. The age of the patient and perforation of colon are not good indicators of survival length of the patient.
2. The extent of disease spread to lymph nodes and areas surrounding the colon are significant. However, they should be studied in conjunction with other factors.
3. The recurrence-free survival length under treatment "Lev+5-FU" in group 2 nearly 100% greater than the corresponding survival length in group 0 and group 1.
4. The survival length is dependent on whether the cancer has spread to other organs near the colon or not, at the time the patient registered for adjuvant therapy.
5. The dataset can however be used to eliminate factors such as age, perforation of colon and gender because they do not appear to have a strong correlation with survival or death.

## VI. LIMITATIONS

The following limitations are relevant to the analysis presented in this paper.

1. The conclusions based on a single trial are usually not sufficient proof of a drug's effectiveness. Therefore, they need to be corroborated by other studies as well.
2. Since the dataset does not clearly differentiate between stage B and stage C colon cancer patients, another dataset which provides this differentiation could be very helpful in understanding the specific stage of cancer for which the combination of Levamisole and Fluorouracil drugs is relatively more effective.

## VII. REFERENCES

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